

Assignment 3

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Sec : 10

Course : CSE 423

Projection

Ans to the Q.no-4

$$P(10, 20, -40)$$

$$\beta = 30^\circ$$

$$\lambda = 0.5 \text{ [Cabinet Projection]}$$

For projection plane at $x=7$

$$x' = 7$$

$$\bar{y} = y + \lambda \cos \beta x$$

$$\bar{z} = z + \lambda \sin \beta x$$

$$w = 1$$

$$\begin{bmatrix} x' \cdot w \\ \bar{y} \cdot w \\ \bar{z} \cdot w \\ w \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 7 \\ 0.5 \cos 30^\circ & 1 & 0 & 0 \\ 0.5 \sin 30^\circ & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x' \cdot w \\ \bar{y} \cdot w \\ \bar{z} \cdot w \\ w \end{bmatrix} = \begin{bmatrix} 7 \\ 24.33 \\ -37.5 \\ 1 \end{bmatrix}$$

$$\therefore P'(7, 24.33, -37.5)$$

(Ans)

2)

Ans to the Ques-7

(a) $P(10, 20, -40)$

For orthographic projection, $\lambda = 0$

For any plane projection,

$$x' = x$$

$$y' = y$$

$$z' = 0$$

$$w = 1$$

$$\begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x' \cdot w \\ y' \cdot w \\ z' \cdot w \\ w \end{bmatrix} = \begin{bmatrix} 10 \\ 20 \\ 0 \\ 1 \end{bmatrix}$$

$$\therefore P'(10, 20, 0)$$

(b) For orthographic projection, $\lambda=0$

For projection plane $y=13$

$$\bar{y}=13$$

$$\bar{z}=z$$

$$\bar{x}=x$$

$$w=1$$

$$\begin{bmatrix} \bar{x}.w \\ \bar{y}.w \\ \bar{z}.w \\ w \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 13 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ -40 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} \bar{x}.w \\ \bar{y}.w \\ \bar{z}.w \\ w \end{bmatrix} = \begin{bmatrix} 10 \\ 13 \\ -40 \\ 1 \end{bmatrix}$$

$$P'(10, 13, -40)$$

(8) Cavalier projection is called full length projection because the lines extending along the receding axis are rendered at their exact, true length without any foreshortening.